Topic 9 – Designing for the cleaner environment

9.1 Cleaner design and the product life cycle

Spec: Material selection – source, quantity, quality, range, recyclability, biodegradability; Manufacture – minimising energy use, simplification of processes, achieving optimum use of materials and components, giving consideration to material form, cost and scale of production; Distribution – efficient use of packaging, reduction of transport, alternatives to fossil fuels; Use – repair versus replacement, energy efficiency, efficiency ratings; Repair and maintenance – standardisation, modular construction, bought in parts; End of life – design for disassembly, recovered material collection, sorting and re-processing methods, energy recovery, environmental implications of disposal to landfill

Key facts

• **Six stages to consider in cleaner design?** Material selection, manufacture, distribution, use, repair and maintenance, end of life.
• **Material selection considerations?** Source (renewable, FSC, recycled, ethical, local), quantity (use less, lightweighting), quality, range (fewer types aids recycling), recyclability, biodegradability.
• **How can manufacture be cleaner?** Minimise energy (efficient machines, renewables, ISO 50001), simplify processes and fewer parts, nest parts to cut waste, recycle offcuts/sprues, choose stock form and scale wisely.
• **How can distribution be cleaner?** Minimal/recyclable packaging, flat-pack and nesting to fit more per vehicle, local manufacture, efficient routes, electric/hydrogen/biofuel vehicles, rail/sea instead of air.
• **Repair vs replacement?** Repair extends life and saves resources; needs available spare parts, modular design, standard fixings, repair manuals – 'right to repair'.
• **Methods of sorting recovered materials?** Hand picking, magnets (ferrous), eddy currents (aluminium), air classification, flotation (density), near-infrared (NIR) optical sorting of plastics, trommels/screens.
• **Environmental implications of landfill?** Wastes resources, methane (greenhouse gas) from decomposition, leachate pollutes soil and water, uses land, visual pollution, landfill tax.
• **Disadvantages of biodegradable polymers?** More expensive, need specific composting conditions (PLA – industrial), may release methane in landfill, contaminate recycling streams, use farmland.
• **Name two end of life design considerations.** Design for disassembly; recyclable and identifiable materials; biodegradability; reuse/repurposing; energy recovery; minimising landfill.

Key terms

Life cycle assessment (LCA)	Analysing the environmental impact of a product at every stage – raw materials, manufacture, distribution, use and end of life ('cradle to grave').
Carbon footprint	The total greenhouse gas emissions caused by a product or activity, measured in kg CO_2 equivalent.
Embodied energy	The total energy used to extract, process, manufacture and transport a material or product.
The 6 Rs	Reduce, Reuse, Recycle, Rethink, Refuse, Repair (sometimes Rot/Re-purpose).
Design for disassembly (DfD)	Designing products so they can be easily taken apart for repair, reuse or recycling – mechanical fixings, fewer material types, labelled parts.
Modular construction	Products built from standard interchangeable modules that can be replaced or upgraded individually.
Energy recovery	Burning non-recyclable waste in energy-from-waste plants to generate electricity and heat.
Upcycling	Creatively reusing unwanted products or materials to make something of higher value.
Energy efficiency rating	A label (e.g. UK/EU A–G energy label) showing how efficiently an appliance uses energy.
Biopolymer	Polymer made from renewable plant sources and/or that biodegrades – e.g. PLA, starch-based polymers, PHB (Biopol).

Exam tip (2023 Q4d, 2021 Q7b): Structure answers around the stages – materials, manufacture, distribution, use, repair, end of life – and give specific actions for each.

9.2 Wider issues of cleaner technologies

Spec: Cost implications to the consumer and manufacturer; Sustainability – designing without jeopardising the potential for people in the future to meet their needs

Key facts

• **Define sustainability (spec wording).** Designing without jeopardising the potential for people in the future to meet their needs.
• **Costs of cleaner technology to manufacturers?** Investment in new equipment and processes, R&D, training, certification, more expensive sustainable materials; but savings from less energy, waste, landfill tax, and better brand image.
• **Costs to consumers?** Higher purchase price (green premium) but lower running costs (energy efficient appliances, EVs), longer life, possible grants/incentives.
• **Three pillars of sustainability?** Environmental, social and economic.
• **How does legislation drive cleaner design?** WEEE, RoHS, packaging regulations, carbon taxes, landfill tax, plastic packaging tax, right-to-repair, ecodesign regulations.

	Manufacturer	Consumer
Extra costs	New equipment and processes, R&D, staff training, certification (ISO 14001), sustainable/recycled materials may cost more, take-back schemes, compliance with legislation	Higher purchase price ("green premium"), may need to change habits (returning products, sorting waste)
Savings / benefits	Lower energy and material use, less waste and landfill tax, avoid carbon taxes and fines, government grants, improved brand image and sales to eco-aware customers, new markets (repair, remanufacture)	Lower running costs (energy-efficient appliances, EVs), longer-lasting and repairable products, grants and incentives, resale/buy-back value

Key terms

Sustainability	Meeting the needs of the present without compromising the ability of future generations to meet their own needs (Brundtland Report, 1987).
Cleaner technology	Processes, products and services that reduce pollution, waste and resource use compared with conventional alternatives.
Triple bottom line	Measuring success by People (social), Planet (environment) and Profit (economic).
Green premium	Extra cost consumers may pay for environmentally friendly products.
Whole-life cost	Total cost of owning a product – purchase price plus running, maintenance and disposal costs.

Exam tip: In discussion questions, mention **both** costs and benefits to **both** the consumer and the manufacturer, then reach a judgement.